

How to (Re)Invent Tait's Method*

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Spring, 2026

1 Introduction

Two of the most important developments in type theory were the invention, by W. W. Tait, of *Tait's Method* for function types, which was later extended by J.-Y. Girard to *Girard's Method* for type quantification, both of which were incorporated into a general theory of *logical relations* for a wide range of type theories. Tait's method continues to be known by its original name, the *computability method*, which interprets types as predicates in the manner developed below.¹

The problem considered by Tait was to prove that β -reduction for the simply typed λ -calculus is *strongly normalizing*, which is usually defined to mean that there are *no* infinite β -reduction sequences starting with a well-typed term. The question considered here is related, but technically much simpler, the termination of a deterministic head reduction strategy for a simply typed λ -calculus. The type system considered here has unit, product, and function types, augmented with a type of *answers*, yes or no, corresponding to the familiar accept or reject distinction for abstract machines.

2 Simple Types

The syntax of the language considered here is given by the following grammar:

$$\begin{aligned} A &::= \text{Ans} \mid 1 \mid A_1 \times A_2 \mid A_1 \rightarrow A_2 \\ M &::= x \mid \text{yes} \mid \text{no} \mid \langle \rangle \mid \langle M_1, M_2 \rangle \mid M \cdot 1 \mid M \cdot 2 \mid \lambda(x.M) \mid \text{ap}(M_1; M_2) \end{aligned}$$

The statics of the language is given in Figure 1. These rules constitute an inductive definition of the judgment $\Gamma \vdash M : A$, and are formulated so as to ensure that the structural properties of entailment are admissible.

Define $\Gamma' \vdash \gamma : \Gamma$ to mean that γ is a finite function defined on variables declared in Γ such that if $\Gamma \vdash x : A$, then $\Gamma' \vdash \gamma(x) : A$. Such a mapping determines a substitution function, $\hat{\gamma}$, on terms that replaces each such x with $\gamma(x)$ in the term.²

Lemma 1 (Substitution). *If $\Gamma \vdash M : A$ and $\Gamma' \vdash \gamma : \Gamma$, then $\Gamma' \vdash \hat{\gamma}(M) : A$.*

The *structural properties* of typing follow immediately:

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¹It can be confusing, at first, that Tait's notion of computability has nothing to do with computability theory!

²This lemma is remarkably difficult to prove in full generality; see Aydemir et al. (2008) for more on this issue.

VAR	YES	NO	UNIT
$\frac{}{\Gamma, x : A \vdash x : A}$	$\frac{}{\Gamma \vdash \text{yes} : \text{Ans}}$	$\frac{}{\Gamma \vdash \text{no} : \text{Ans}}$	$\frac{}{\Gamma \vdash \langle \rangle : 1}$
PAIR	LFT	RHT	
$\frac{\Gamma \vdash M_1 : A_1 \quad \Gamma \vdash M_2 : A_2}{\Gamma \vdash \langle M_1, M_2 \rangle : A_1 \times A_2}$	$\frac{\Gamma \vdash M : A_1 \times A_2}{\Gamma \vdash M \cdot 1 : A_1}$	$\frac{\Gamma \vdash M : A_1 \times A_2}{\Gamma \vdash M \cdot 2 : A_2}$	
LAM	APP		
$\frac{\Gamma, x : A_1 \vdash M_2 : A_2}{\Gamma \vdash \lambda(x.M_2) : A_1 \rightarrow A_2}$	$\frac{\Gamma \vdash M_1 : A_2 \rightarrow A \quad \Gamma \vdash M_2 : A_2}{\Gamma \vdash \text{ap}(M_1; M_2) : A}$		

Figure 1: Typed λ -Calculus Statics

YES	NO	UNIT	PAIR	LAM	LFT
$\frac{}{\text{yes val}}$	$\frac{}{\text{no val}}$	$\frac{}{\langle \rangle \text{ val}}$	$\frac{}{\langle M_1, M_2 \rangle \text{ val}}$	$\frac{}{\lambda(x.M_2) \text{ val}}$	$\frac{M \mapsto M'}{M \cdot 1 \mapsto M' \cdot 1}$
RHT	LFT-PAIR	RHT-PAIR	APP		
$\frac{M \mapsto M'}{M \cdot 2 \mapsto M' \cdot 2}$	$\frac{}{\langle M_1, M_2 \rangle \cdot 1 \mapsto M_1}$	$\frac{}{\langle M_1, M_2 \rangle \cdot 2 \mapsto M_2}$	$\frac{M_1 \mapsto M'_1}{\text{ap}(M_1; M_2) \mapsto \text{ap}(M'_1; M_2)}$		
	APP-LAM				
	$\frac{}{\text{ap}(\lambda(x.M); M_2) \mapsto [M_2/x]M}$				

Figure 2: Typed λ -Calculus Dynamics

1. Reflexivity: $x : A \vdash x : A$. This is an instance of the reflexivity rule.
2. Transitivity: If $\Gamma, x : A \vdash N : B$, then $\Gamma \vdash M : A$ implies $\Gamma \vdash [M/x]N : B$.
3. Weakening: If $\Gamma \vdash N : B$, then $\Gamma, x : A \vdash N : B$, where x is not declared in Γ .
4. Contraction: If $\Gamma, x_1 : A, x_2 : A \vdash N : B$, then $\Gamma, x : A \vdash [x, x/x_1, x_2]N : B$.

Exercise 1 (Difficult). *Attempt to prove Lemma 1 by induction on the derivation of $\Gamma \vdash M : A$. There is one spot where your proof is sure to break down. Isolate that spot, and consider other ways to recover the intended result.*

The dynamics is given by a transition system $M \mapsto M'$ between closed λ -terms of some type. Any closed typed term is a valid initial state. Final states are defined along with transition in Figure 2.

Theorem 2 (Preservation). *If $M : A$ and $M \mapsto M'$, then $M' : A$.*

Proof. By induction on transition. □

3 Termination Proof

The goal is to prove termination for terms of observable type:

Theorem 3 (Termination). *If $M : \text{Ans}$, then either $M \mapsto^* \text{yes}$ or $M \mapsto^* \text{no}$.*

That is, any complete program either accepts or rejects.

Given the statement of the theorem, practically the only move available is to proceed by induction on typing. Let us consider some cases.

VAR Does not apply to closed terms.

YES Immediate, as yes val.

NO Immediate, as no val.

UNIT Does not apply, not of type Ans.

PAIR Does not apply, not of type Ans.

LFT By induction, $um \dots$

RHT By induction, $um \dots$

LAM Does not apply, not of type Ans.

APP By induction applied to the first premise, $um \dots$

All cases are trivial, or completely unclear.

Well, because the subterms of a term of type Ans need not have type Ans, it seems clear that it is necessary to strengthen the theorem to say something about terms of any type.

Lemma 4. *If $M : A$, then there exists N such that $N \text{ val}$ and $M \mapsto^* N$.*

The lemma suffices for the theorem because of the definition of finality for terms of type Ans. Let us consider the proof of this lemma.

VAR Does not apply to closed terms.

YES Immediate, as yes val.

NO Immediate, as no val.

UNIT Immediate, as $\langle \rangle$ val.

PAIR Immediate, as $\langle M_1, M_2 \rangle \text{ val}$.

LFT By induction there exists N such that $N \text{ val}$ and $M \mapsto^* N$. By preservation and the definition of finality N must be of the form $\langle N_1, N_2 \rangle$. By the definition of transition

$$M \cdot 1 \mapsto^* \langle N_1, N_2 \rangle \cdot 1 \mapsto N_1.$$

But now what?

RHT Analogous, what to do with N_2 ?

LAM Immediate, as $\lambda(x.M_2) \text{ val}$.

APP By induction applied to the first premise there exists N_1 such that $N_1 \text{ val}$ and $M_1 \mapsto^* N_1$. By preservation and the definition of finality N_1 must have the form $\lambda(x.M)$. By the definition of transition

$$\text{ap}(M_1; M_2) \mapsto^* \text{ap}(\lambda(x.M); M_2) \mapsto [M_2/x]M.$$

But now what?

In the projection cases the components of the pair are general terms about which nothing is known. In the application case the value of the first argument is a λ -abstraction whose body is an open term (with free variable x) about which nothing is known. This suggests strengthening the lemma by proving a property called *hereditary termination*, which is stronger than mere termination. It should have the following characteristics in order to push through the proof of the strengthened lemma below:

1. A hereditarily terminating expression of type 1 should be terminating, and hence transition to $\langle \rangle$.
2. A hereditarily terminating expression of type Ans should be terminating, and hence transition to either yes or no.
3. A hereditarily terminating expression of type $A_1 \times A_2$ should terminate with a pair $\langle N_1, N_2 \rangle$ such that both N_1 and N_2 are hereditarily terminating.
4. A hereditarily terminating expression of type $A_2 \rightarrow A$ should terminate with a function $\lambda(x.M)$ such that if M_2 is hereditarily terminating of type A_2 , then $[M_2/x]M$ should be hereditarily terminating at type A .

These conditions constitute a *definition* of the property M is *hereditarily terminating at type A*, which is defined for closed $M : A$. The first two cases are given outright; the others rely on hereditary termination at constituent types of a compound type. *Thus, hereditary termination at a type is defined by induction on the structure of the type.*³

Lemma 5. *If $M : A$, then $HT_A(M)$.*

The proof proceeds as before by induction on typing. The cases for the constants are immediate by the definition of hereditary termination at base type.

The problematic elimination cases use the definition of hereditary termination, along with an additional property, called *head expansion*. Before stating it, let us see how it arises. Consider the rule LFT

³For reference the type-indexed family of predicates, $HT_A(M)$, defining hereditary termination is given in Figure 3.

$$\begin{aligned}
\text{HT}_1(M) &\text{ iff } M \mapsto^* \langle \rangle \\
\text{HT}_{\text{Ans}}(M) &\text{ iff } M \mapsto^* \text{yes} \text{ or } M \mapsto^* \text{no} \\
\text{HT}_{A_1 \times A_2}(M) &\text{ iff } M \mapsto^* \langle M_1, M_2 \rangle \text{ and } \text{HT}_{A_1}(M_1) \text{ and } \text{HT}_{A_2}(M_2) \\
\text{HT}_{A_1 \rightarrow A_2}(M) &\text{ iff } M \mapsto^* \lambda(x.M_2) \text{ and } \text{HT}_{A_1}(M_1) \text{ implies } \text{HT}_{A_2}([M_1/x]M_2) \\
\text{HT}_\Gamma^*(\gamma) &\text{ iff } \text{HT}_A(\gamma(x)) \text{ for all } x : A \in \Gamma
\end{aligned}$$

Figure 3: Hereditary Termination, $\text{HT}_A(M)$

once again. By induction on the premise of the rule, $\text{HT}_{A_1 \times A_2}(M)$. By the definition of hereditary termination $M \mapsto^* \langle M_1, M_2 \rangle$ and $\text{HT}_{A_1}(M_1)$. To show $\text{HT}_{A_1}(M \cdot 1)$, observe that

$$M \cdot 1 \mapsto^* \langle M_1, M_2 \rangle \cdot 1 \mapsto M_1.$$

To complete the proof it suffices to show that hereditary termination is closed under “reverse execution”.

Lemma 6 (Head Expansion). *If $\text{HT}_A(M)$ and $M' \mapsto M$, then $\text{HT}_A(M')$.*

Proof. Immediate, because the definition of hereditary termination is defined in terms of the evaluation behavior of terms. $\square$

This completes the proof for the rule LFT; rules RHT and APP are handled similarly.

What about the pair and function cases?

PAIR By induction M_1 is hereditarily terminating at A_1 and M_2 is hereditarily terminating at type A_2 ; the goal is to show that $\langle M_1, M_2 \rangle$ is hereditarily terminating at type $A_1 \times A_2$. A pair is already a value (final state), so an appeal to the inductive hypothesis suffices to finish the proof.

LAM To show that $\lambda(x.M_2)$ is hereditarily terminating at $A_1 \rightarrow A_2$, show that whenever M_1 is hereditarily terminating at A_1 , then $[M_1/x]M_2$ is hereditarily terminating at A_2 . But what to do?

The problem now is that in the function case there is no inductive hypothesis available to give us the necessary information about the *open* term M , which has one free variable, x , in it. The lemma must be strengthened once more to account for open terms, even though the desired property applies only to closed terms.

The judgment $\Gamma \gg M \in A$ means that if $\text{HT}_\Gamma^*(\gamma)$, then $\text{HT}_A(\hat{\gamma}(M))$.

Theorem 7. *If $\Gamma \vdash M : A$, then $\Gamma \gg M \in A$.*

Proof. The proof is by induction on typing derivations. First, the variable rule now comes into play, and is handled by assumption. Specifically, to prove that $\Gamma, x : A \gg x \in A$, suppose that $\text{HT}_\Gamma^*(\gamma)$ and $\gamma(x) = M$ such that $\text{HT}_A(M)$, so that $\text{HT}_{\Gamma, x:A}(\gamma, x \hookrightarrow M)$, with the goal to show $\text{HT}_A(\hat{\gamma}(x))$. But this follows immediately from the definition substitution, $\hat{\gamma}(x) = \gamma(x) =$, and we have $\text{HT}_A(M)$

by assumption. Second, consider the typing rule for λ -abstractions given in Figure 1. The goal is to show that if $HT_{\Gamma}^*(\gamma)$, then $HT_{A_1 \rightarrow A_2}(\hat{\gamma}(\lambda(M_2)))$. To this end, assume that $HT_{\Gamma}^*(\gamma)$, and assume further that $HT_{A_1}(M_1)$, with the goal to show that $HT_{A_2}([M_1/x]\hat{\gamma}(M_2))$. Let $\gamma' \triangleq \gamma, x \hookrightarrow M_1$, and note that $HT_{\Gamma, x:A_1}(\gamma')$, so that by induction $HT_{A_2}(\hat{\gamma}'(M_2))$. But $\hat{\gamma}'(M_2) = [M_1/x]\hat{\gamma}(M_2)$, and thus the desired result. The other cases are essentially the same as sketched above, albeit applying $\hat{\gamma}$ to the constituent terms in the premises and conclusion of each rule. $\square$

Exercise 2. If termination is required only for closed programs of answer type, and not for higher types, then a “negative” formulation of hereditary termination is sensible:

$$\begin{aligned} HT_{A_1 \times A_2}(M) &\text{ iff } HT_{A_1}(M \cdot 1) \text{ and } HT_{A_2}(M \cdot 2) \\ HT_{A_1 \rightarrow A_2}(M) &\text{ iff } HT_{A_1}(M_1) \text{ implies } HT_{A_2}(\text{ap}(M; M_1)) \end{aligned}$$

Re-prove the termination theorem using this revised definition of hereditary termination at product and function types.

Exercise 3. Finite sums, given by the empty type 0 and the binary sum, $A_1 + A_2$, require a “positive” formulation of hereditary termination:

$$\begin{aligned} HT_0(M) &\text{ iff (never)} \\ HT_{A_1 + A_2}(M) &\text{ iff } M \mapsto^* 1 \cdot M_1 \text{ and } HT_{A_1}(M_1), \text{ or} \\ &\quad M \mapsto^* 2 \cdot M_2 \text{ and } HT_{A_2}(M_2). \end{aligned}$$

Extend the proof of termination to account for sum types based on these definitions. What would be a “negative” formulation of sum types? What goes wrong?

Exercises 4 and 5 following concern to the statics and dynamics of the types of natural and co-natural numbers given in Figure 4.

Exercise 4. Define hereditary termination at type Nat as the strongest property $\mathcal{P}$ of $M : \text{Nat}$ such that if either

1. $M \mapsto^* \text{zero}$, or
2. $M \mapsto^* \text{succ}(N)$ with $\mathcal{P}(N)$,

then $\mathcal{P}(M)$. Extend the proof of termination given above to account for this type.

Exercise 5. Define hereditary termination at type Conat to be the weakest property $\mathcal{P}$ of $M : \text{Conat}$ such that if $\mathcal{P}(M)$, then either

1. $\text{out}(M) \mapsto^* 1 \cdot N$ with $HT_1(N)$, or
2. $\text{out}(M) \mapsto^* 2 \cdot N$ with $\mathcal{P}(N)$.

Extend the termination proof to account for the type Conat of co-natural numbers.

$$\begin{array}{c}
\begin{array}{ccc}
\text{NAT-I-Z} & \text{NAT-I-S} & \text{NAT-E} \\
\frac{}{\Gamma \vdash \text{zero} : \text{Nat}} & \frac{\Gamma \vdash M : \text{Nat}}{\Gamma \vdash \text{succ}(M) : \text{Nat}} & \frac{\Gamma \vdash M : \text{Nat} \quad \Gamma \vdash M_0 : A \quad \Gamma, x : A \vdash M_1 : A}{\Gamma \vdash \text{rec}(M; M_0; x.M_1) : A} \\
\text{ZERO-VAL} & \text{SUCC-VAL} & \text{REC-STEP} \\
\frac{}{\text{zero val}} & \frac{}{\text{succ}(M) \text{ val}} & \frac{M \mapsto M'}{\text{rec}(M; M_0; x.M_1) \mapsto \text{rec}(M_0; x; M_1.)} \\
\text{REC-STEP-Z} & \text{REC-STEP-S} & \\
\frac{}{\text{rec}(\text{zero}; M_0; x.M_1) \mapsto M_0} & \frac{}{\text{rec}(\text{succ}(M); M_0; x.M_1) \mapsto [\text{rec}(M; M_0; x.M_1)/x]M_1} & \\
\text{CONAT-E} & \text{CONAT-I} & \text{PRED-STEP} \\
\frac{\Gamma \vdash M : \text{Conat}}{\Gamma \vdash \text{out}(M) : 1 + \text{Conat}} & \frac{\Gamma \vdash M : A \quad \Gamma, x : A \vdash N : 1 + A}{\Gamma \vdash \text{gen}(M; x.N) : \text{Conat}} & \frac{M \mapsto M'}{\text{out}(M) \mapsto \text{out}(M')} \\
\text{PRED-GEN} & & \\
\frac{}{\text{out}(\text{gen}(M; x.N)) \mapsto \text{case } [M/x]N \{ _ . 1 \cdot \langle \rangle \mid y. \text{gen}(y; x.N) \}} & &
\end{array}
\end{array}$$

Figure 4: Natural and Co-Natural Numbers

A Solutions

Natural Numbers

The introductory rules follow directly from the definition. The eliminatory rule is proved using the minimality of hereditary termination. The premises of the rule give rise to these three facts:

1. $\text{HT}_{\text{Nat}}(\widehat{M})$;
2. $\text{HT}_A(\widehat{M}_0)$;
3. if $\text{HT}_A(N)$, then $\text{HT}_A([N/x]\widehat{M}_1)$.

For convenience, define $R(-) \triangleq \text{rec}(-; \widehat{M}_0; x.\widehat{M}_1)$; the goal is to show that $\text{HT}_A(R(\widehat{M}))$. It suffices to show that the property $\mathcal{P}(-) \triangleq \text{HT}_A(R(-))$ satisfies the following two conditions:

1. If $N \mapsto^* \text{zero}$, then $R(N) \mapsto^* \widehat{M}_0$. It follows that $\mathcal{P}(N)$ by head expansion.
2. If $N \mapsto^* \text{succ}(N')$ with $\mathcal{P}(N')$, that is, $\text{HT}_A(R(N'))$. Now $R(N) \mapsto^* [R(N')/x]\widehat{M}_1$, and so by induction $\text{HT}_A([R(N')/x]\widehat{M}_1)$, from which $\mathcal{P}(N)$ follows by head expansion.

Therefore, by the first inductive assumption, $\mathcal{P}(\widehat{M})$, as required.

Conatural Numbers

The eliminatory rule follows directly from the definition. For the introductory rule, the goal is to show that $\text{HT}_{\text{Conat}}(\text{gen}(\widehat{M}; x.\widehat{N}))$, given the inductive hypotheses

1. $\text{HT}_A(\widehat{M})$, and
2. if $\text{HT}_A(P)$, then $\text{HT}_{1+A}([P/x]\widehat{N})$.

The proof is by co-induction. Writing $G(-) \triangleq \text{gen}(-; x.\widehat{N})$, it suffices to find a property $\mathcal{P}$ of closed terms of type Conat that is consistent with the defining conditions for Conat , and for which $\mathcal{P}(G(\widehat{M}))$, for then $\text{HT}_{\text{Conat}}(G(\widehat{M}))$, as desired. Let $\mathcal{P}(M)$ be defined by

$$M \mapsto^* G(P) \text{ for some } P \text{ s.t. } \text{HT}_A(P).$$

Observe that $G(\widehat{M}) \text{ val}$, and so by the first inductive hypothesis $\mathcal{P}(G(\widehat{M}))$, as required for the co-induction.

To show that $\mathcal{P}$ is consistent, suppose that $\mathcal{P}(M)$, which is to say that $M \mapsto^* G(P)$ for some $\text{HT}_A(P)$, with the goal to show that either

1. $\text{out}(M) \mapsto^* 1 \cdot M'$ with $\text{HT}_1(M')$, or
2. $\text{out}(M) \mapsto^* 2 \cdot M'$ with $\mathcal{P}(M')$.

Now

$$\text{out}(M) \mapsto^* \text{out}(G(P)) \mapsto \text{case } [P/x]\widehat{N} \{ _ . 1 \cdot \langle \rangle \mid y . 2 \cdot \text{gen}(y; x.\widehat{N}) \}.$$

By the second inductive hypothesis $\text{HT}_{1+A}([P/x]\widehat{N})$, from which the result follows by case analysis.

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